Suyoga Gaikwad- TE_A_40 Practical-4

/* index.html */

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Fancy Calculator</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="calculator">
        <input type="text" id="display" disabled placeholder="0">
        <br>
        <div class="buttons">
            <button class="operator special" onclick="clearDisplay()">C</button>
            <button class="operator special" onclick="calculateSquareRoot()">√</button>
            <button class="operator special" onclick="appendOperator('^2')">x²</button>
            <button class="operator special" onclick="appendOperator('/')">/</button>

            <button class="number" onclick="appendNumber(7)">7</button>
            <button class="number" onclick="appendNumber(8)">8</button>
            <button class="number" onclick="appendNumber(9)">9</button>
            <button class="operator special" onclick="appendOperator('*')">*</button>

            <button class="number" onclick="appendNumber(4)">4</button>
            <button class="number" onclick="appendNumber(5)">5</button>
            <button class="number" onclick="appendNumber(6)">6</button>
            <button class="operator special" onclick="appendOperator('-')">-</button>

            <button class="number" onclick="appendNumber(1)">1</button>
            <button class="number" onclick="appendNumber(2)">2</button>
            <button class="number" onclick="appendNumber(3)">3</button>
            <button class="operator special" onclick="appendOperator('+')">+</button>

            <button class="number" onclick="appendNumber(0)">0</button>
            <button class="operator special" onclick="appendOperator('.')">.</button>
            <button class="equal" onclick="calculateResult()">=</button>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

/* style.css */

```css
body {
    font-family: Arial, sans-serif;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    background: linear-gradient(to right, #ff9a9e, #fad0c4);
}
.calculator {
```

```
    background: #FBF8EF;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0,
0, 0, 0.2);
    text-align: center;
}
input {
    width: 100%;
    height: 50px;
    font-size: 24px;
    text-align: right;
    margin-bottom: 10px;
    border: 3px solid #FF7043;
    border-radius: 5px;
    background: #f3f3f3;
    padding-right: 10px;
}
.buttons {
    display: grid;
    grid-template-columns: repeat(4,
1fr);
    gap: 10px;
}
button {
    width: 100%;
    height: 60px;
    font-size: 20px;
    border: none;
    border-radius: 10px;
    cursor: pointer;
    transition: all 0.3s;
}
button:hover {
    background: #B4EBE6;
}
.number {
    background: #80CBC4;
}
.operator.special {
    background: #FF4081;
}
.operator.special:hover {
    background: #FF80A5;
}
.equal {
    background: rgb(255, 180, 51);
}
.equal:hover {
    background: rgba(234, 222, 55,
0.8);
}
```

```
/* script.js */
function appendNumber(num) {
    document.getElementById("display
").value += num;
}

function appendOperator(op) {
    let display =
document.getElementById("display");
    if (display.value === "") {
        alert("Enter a number
first");
        return;
    }
    display.value += op;
}

function calculateSquareRoot() {
    let display =
document.getElementById("display");
    let number =
parseFloat(display.value);
    if (isNaN(number) || number < 0)
{
        alert("Invalid input for
square root!");
        display.value = "";
    } else {
        display.value =
Math.sqrt(number);
    }
}

function calculateResult() {
    let display =
document.getElementById("display").v
alue;
    try {
        if (display.includes("^2"))
{
            let number =
parseFloat(display.split("^2")[0]);
```

```
            if (isNaN(number)) throw
"Invalid input";
            document.getElementById(
"display").value = Math.pow(number,
2);
        } else {
            document.getElementById(
"display").value = eval(display);
        }
    } catch {
        alert("Invalid
expression!");
        document.getElementById("dis
play").value = "";
    }
}

function clearDisplay() {
    document.getElementById("display
").value = "";
}
```